

Name	Priority	Demo	Notes
Research topic 1	2	Have a citations list with references to refer to when working on the class diagrams. Have a small test class diagram ready created on our chosen tool for familiarisation.	Considering using Wondershare Edraw max to draw class diagrams
Research topic 2	2	Have a citations list with references and notes to refer to when working on the program (eg. all useful git commands)	References so far include: https://git-scm.com/cheat-sheet
Research java	1	Have notes and references to refer to regarding the swing and any other highly relevant java topics.	References so far include: https://enos.itcollege.ee/~jpoial/allalaadimised/reading/Advanced-java.pdf , https://youtube.com/playlist?list=PL_QPQmz5C6WUF-pOQDsbsKbaBZqXj4qSq&si=rUQwaTS_zMB73nsg
Render background	1	Start the application and a window of the correct size with appropriately sized panels for each feature of the game to be displayed (lanes, inputs, hearts, bingo).	Decide on final static constants for window sizes and panel sizes and fix overall layout for the game
Render lanes	1	Start the application and 4 simple lanes appear, each its own object with operator at the bottom of the lane.	
Render blocks	1	Start the application and blocks are animated on each lane at random intervals with random numbers (within constraints) that fall from the top to the bottom of the lane panel.	See mathematical logic for constraints
Build mathematical logic	1	For each block and lane, appropriate numbers appear on the block (eg. only multiples of 3 for x3 lanes) and the block number, lane operation and correct answer is logged in the terminal.	At start of game, lane multiplicand, divisor, summand, subtrahend are initialised randomly from [2, 9]. Multiplication lane must only have blocks that are multiples of the multiplicand (up to x10), and all lanes must be such that the correct answer is less than 100 (for level 1, 2, 3).
Render input boxes	1	An input box is added for each lane. Each can be focused (can be typed into) by clicking on it with the mouse, switching with arrow keys or by pressing tab to go to the lane with the closest block.	
Combine mathematical logic and input boxes	1	Each input box accepts an input and checks the answer based on the correct answer specified in the mathematical logic. If it is correct, the corresponding block disappears. If it is wrong, the input box turns red or shakes.	
Render hearts (subtract)	1	Heart panel has 3 hearts at the start of the game, any block that reaches the bottom of the lane panel before the player inputs the correct answer reduces one heart.	
Render hearts (add)	3	Anytime all lane blocks are completed before the halfway point, one heart is added.	
Render score	1	Each completed block adds (level * 100) score.	
End game	1	Once the number of hearts is zero, end the game and display the final score.	
Level 2	2	Slightly increase the speed of the falling blocks. Increase the level by 1 once 16 blocks have been completed.	
Level 3	2	Block numbers are greater to increase difficulty.	Constraints on the mathematical logic are modified. Multiplication lane can have multiples up to x 12, division lane has correct answer up to 144 (ie. 12 x 12 times table), addition and subtraction lanes can have correct answers less than 200.
Reinitialise lanes	3	At random intervals (every ~20-30 blocks completed), one of the multiplicand, divisor, summand or subtrahend of the lanes are randomly reinitialised from [2, 9]. When this happens, the changed lane is briefly highlighted in a dark red colour.	
Render red blocks	3	At random intervals (every ~20-30 blocks), a block is coloured red. If these red blocks fall before the correct answer is inputted, all remaining hearts are gone regardless of how many hearts are remaining.	
Render level clearing	2	When each level is cleared, blocks stop falling for ~5 seconds and a big text with the new level appears in the middle of the screen during this time.	
Infinite levels	3	Block numbers increase in difficulty as a function of the level.	Constraints on the mathematical logic are modified. Multiplication lane can have multiples up to x (9 + level), division lane has correct answer up to (9 + level) * 2 (ie. (9 + level) x (9 + level) times table), addition and subtraction lanes can have correct answers less than (level - 1) * 100.
(Mode 2) Render bingo panel	3	When mode 2 is entered, a separate pannel appears besides the lanes that has numbers from [1, 10]	
(Mode 2) Two arguments input	3	The lanes in this mode accept 2 numbers (separated by a space) and they only display operations (addition, subtraction, multiplication, division).	The operation is applied to the 2 inputs and the correct answer is such that it results in the block number.
(Mode 2) Render hearts (add)	3	When the additional pannel is completed, a heart is added to the heart meter	
(Mode 2) Level 3	3	Add 11 and 12 to bingo panel	
(Mode 2) Infinite levels	3	Add (9 + level) number to the bingo panel, for example, on level 4, 13 is added to the bingo panel.	
Menu screen	3	When application is started, a menu screen with the name of the game, how to play and mode selection appears. Selecting a mode starts the game in that mode.	
(Design) Background	3	During the game, the background is black with a faded grey large number at the centre of the screen displaying the level.	
(Design) Block	4	The blocks are visually appealing and have animations (shadows, rounded corners etc.)	
(Design) Fonts, Colours	3	Application looks more appealing with appropriate colours and fonts.	Fonts to be taken from fonts.google.com for copyright-free fonts.
(Design) Sound effects	3	Significant events in the game (block hitting bottom of lane, blocks disappearing, wrong answer, levelling up, gaining a heart, game ending) are accompanied with appropriate sound effects.	Sound effects to be created on Logic Pro (own creation).
(Design) Background Music	4	During the game, suitable background music is played. Whenever there is a level up, there could be a key change.	Background music created on Logic Pro (own creation).
(Design) Dark theme	5	Menu has a dark theme option that when selected, changes the whole application to dark theme	
(Design) Options	5	Menu has further options such as volume of sound effects and background music	
(Mode 3)	5	Mode 3 is conceptualised and implemented somewhat as described on the right. However, this mode is not fully conceptualised at the present moment.	4 lanes but each lane has an operation only (+, *, /, -) At the very bottom of the screen, a number is displayed that changes at random intervals of time. The player must input the number on the correct lane such that the operation of the lane can be performed on the number at the bottom of the screen and on the inputted number to give the block answer number as a result.
Name of the game	3	Have a unique and appealing name for the game.	Same heart, levels etc. concepts as mode 1.

Learning Goals:

- Apply concepts during first weeks in larger project context
- Independently learn 2 different topics
- Learn Java's swing to build UI

Chosen topics:

1. UML Class Diagrams
2. Version control using git

MODE 1 Screen

MODE 2 Screen